

KIMBERLY K. BROUGHTON

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Software Engineer with over 17 years of experience designing, developing, and modernizing enterprise software solutions. Proven expertise in Python, JavaScript, automation, APIs, and database design across scientific research, healthcare, publishing, and government environments. Recognized NASA Inventor with a history of delivering scalable applications, reducing technical debt, automating complex workflows, and collaborating directly with stakeholders to transform business requirements into reliable software solutions.

SIGNIFICANT ACHIEVEMENTS

Sole architect and lead developer of NASA Earthdata Pub Overview and Forms applications being used by all DAACs for all Data Producers. Official NASA 'Inventor' of 'Earthdata Pub' (NASA Case No. GSC-18, 612-1)

- Architecture
- Development
- Cloud migration
- Adoption across NASA DAACs
- APIs
- Dashboard modernization
- Design decisions
- Authentication
- Technical leadership

Designed and implemented administrative software system to replace data processing department 02/2018

UI / UX Certified in User Experience – NN/g Nielsen Norman Group 12/2020

Contributed to ONC Stage 1 and Stage 2 Electronic Health Record Certification: Electronic Health Record Software Certification - 1 of 3 developers assigned

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, GraphQL, SQLAlchemy, C#, C++, Java, Perl, PHP, HTML5, CSS3, XML, Shell, AWK, Visual Basic, VB.NET

Frameworks: Flask, Vue.js, React, Angular, Bootstrap, jQuery, Mustache, Jinja2, Cypress, Webapp2

Cloud & DevOps: AWS Cloud, Docker, Kubernetes, Git, GitLab, GitLab CI/CD Pipelines, GitLab Runners, YAML, Automation Scripting

Databases: PostgreSQL, Oracle, Microsoft SQL Server, MySQL, Informix, Cache, c-tree, Microsoft Access

Development Tools: VS Code, DBVisualizer, Oracle Forms & Reports, Crystal Reports, Jasper Reports, Visual Studio, Swagger, SpectaQL, Sphinx, Vuese

Content Management: Drupal, Joomla, WordPress

AI Development Tools: DUO AI, GitHub Copilot

Operating Systems: Linux, Microsoft Windows

CORE COMPETENCIES

- Software Architecture
- Full Stack Development
- Enterprise Applications
- REST APIs
- Database Design
- Automation
- System Integration
- Technical Leadership
- Application Modernization
- Technical Documentation
- User Experience
- Agile Development

EXPERIENCE

ORNL DAAC – Oak Ridge, TN

04/18 – 08/26

Systems Architect / Application Developer

Developing in: AWS Cloud, Python, JavaScript (above frameworks and DevOps), SQLAlchemy, GraphQL, Perl, HTML, CSS, SQL, Drupal, DBVisualizer, VSCode

- Repurposed NASA's Cumulus Dashboard to create NASA's Earthdata Pub Dashboard for use of my Overview and Forms applications (Vue, React). The public version of the Earthdata Pub Suite of Products can only be used by NASA and data producers via login and password
- Co-developed the Earthdata Pub REST API powering multiple NASA applications
- Assisted in refactor of web toolkit used for evaluating hydropower
- Implemented a new modularized web application called MyWorkSpace as a new data order system to improve users download options

- Was 1 of 3 developers who implemented a new Ingest Dashboard for all ingest related tasks specific to the ORNL DAAC. MVP now out. This is intended to interface with Earthdata Pub to perform Quality Assurance tasks and full automation
- Designed automation scripts that reduced manual processing time by 80%
- Refactored sites and scripts from perl to python, using an efficient templating system
- Implemented automated unit and end-to-end testing, improving deployment confidence and reducing regression defects
- The importing and exporting of metadata in multiple formats
- Modernized and maintained the ORNL DAAC web platform while supporting migration to NASA's cloud-hosted Earthdata environment, improving maintainability and supporting future platform growth

COSBY HARRISON COMPANY (Formerly TAP Publishing Company) – Crossville, TN

10/16 – 04/18

System Architect / Web Application Developer / Web Developer

Developed in: Python, JavaScript, Perl, DOS, AutoHotKey, HTML, CSS, SQL, Oracle, Informix, Shell, C++

- Designed and implemented administrative software system to replace data processing department
- Streamlined and automated publishing workflows of processes reducing hours of manual work to a single button click while virtually eliminating all errors improving company costs
- Importing and Exporting of data in multiple formats for data standardization
- Automated tasks via scripts in multiple environments in multiple languages
- Performing maintenance, feature and framework improvements of custom web sites, publication web sites and company sites

LIBERA, INC. – Knoxville, TN

10/15 – 07/16

Web Application Programmer

Developed in: VB.NET/ASP.NET, XML, C#, TSQL, System7, SQL, Visual Studio 2015

- Performed Vocational Rehabilitation program updates affecting all agencies statewide
- Created custom queries and scripts for agency end users
- Unit testing Vocational Rehabilitation program updates for developers statewide
- Prepared reference documentation for users by writing operating instructions

RMB, INC. – Knoxville, TN

03/15 – 07/15

FACS Application Programmer

Developed in: FACS, VBScript, AutoHotKey, Jasper Reports, Crystal Reports

- Assisted in maintaining database operations at peak performance, protect secured information, and maintain integrity of data
- Created daily jobs and import / export routines to route in and out a large amount of data
- Created custom queries and reports for the purpose of providing timely and accurate information to end users
- Prepared reference for users by writing operating instructions.

QRS, INC. – Knoxville, TN

05/09 – 02/15

EHR Developer / Software Developer

Developed in: Javascript, HTML, CSS, PHP, Python, Perl, Linux, MySQL, MSSQL, c-tree

- Designed Standard Form Sets / Notes according to doctor specialty
- Worked with doctors directly to determine practice needs
- Indirectly managed other EHR programmers
- 2nd Tier Product Support
- Documented and created instructional videos to train clients on Meaningful Use
- Performed HL7 Interface design and implementation as well as data conversions based on client needs
- Designed interfaces and custom programs based on client needs both on back-end and front-end

EDUCATION

UI / UX Certification – NN/g Norman Group

12/20

C# CERTIFICATION – Knoxville, TN

12/17

PELLISSIPPI TECHNICAL COMMUNITY COLLEGE – Knoxville, TN

05/07 – 05/09

A.A.S. Computer Programming

A.A.S. Database Design and Development

- GPA at 3.60
- Dean's List for Fall 2008, Spring 2007